Project 6:-Maven Selenium Project – Firefox Browser

Step 1: Create a Maven Project

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenFirefoxApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~$ cd devops
harith@harith-VirtualBox:~/devops$ cd test
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenFirefoxApp -DarchetypeArtifactId=maven-ar
chetype-quickstart -DinteractiveMode=false
Project Structure:
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
```

Project Structure:

```
harith@harith-VirtualBox:~/devops/test$ tree
.
├── MyMavenApp
│   ├── pom.xml
│   ├── src
│   │   ├── main
│   │   │   └── java
│   │   │       └── com
│   │   │           └── example
│   │   │               └── App.java
│   │   └── test
│   │       └── java
│   │           └── com
│   │               └── example
│   │                   └── AppTest.java
│   └── target
│       ├── classes
│       │   └── com
│       │       └── example
│       │           └── App.class
│       ├── generated-sources
│       │   └── annotations
```

Step 2: Install Firefox Browser

firefox –version

if not installed: sudo apt install firefox

Step 3: Install GeckoDriver

wget https://github.com/mozilla/geckodriver/releases/latest/download/geckodriver-v0.34.0-linux64.tar.gz

tar -xvzf geckodriver-v0.34.0-linux64.tar.gz

sudo mv geckodriver /usr/local/bin/

sudo chmod +x /usr/local/bin/geckodriver

geckodriver --version

Step 4: Add Selenium Dependency in pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>
<artifactId>MyMavenFirefoxApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>


<name>MyMavenFirefoxApp</name>
<dependencies>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<build>
<plugins>
```

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.11.0</version>

<configuration>

<source>11</source>

<target>11</target>

</configuration>

</plugin>

<plugin>

<groupId>org.codehaus.mojo</groupId>

<artifactId>exec-maven-plugin</artifactId>

<version>3.1.0</version>

<configuration>

<mainClass>com.example.App</mainClass>

</configuration>

</plugin>

</plugins>

</build>

</project>

Step 5: Inspect Elements in the Websites

Steps:

1. Open the website.

2. Right-click on the webpage.
3. Click **Inspect**.
4. Dentify the required HTML elements.

Step 6: Create App.java

Location:

src/main/java/com/example/App.java

Java Code:

```
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");

        driver.findElement(By.id("user-name")).sendKeys("standard_user");

        driver.findElement(By.id("password")).sendKeys("secret_sauce");

        driver.findElement(By.id("login-button")).click();

        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://automationexercise.com/products");
```

```
        driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");

        driver.findElement(By.id("submit_search")).click();

        WebElement product = wait.until(
            ExpectedConditions.visibilityOfElementLocated(
                By.cssSelector("a[data-product-id='2']")
            )
        );

        ((org.openqa.selenium.JavascriptExecutor) driver)
            .executeScript("arguments[0].scrollIntoView({block:'center'});", product);

        ((org.openqa.selenium.JavascriptExecutor) driver)
            .executeScript("arguments[0].click();", product);

        WebElement viewCart = wait.until(
            ExpectedConditions.elementToBeClickable(
                By.cssSelector("#cartModal a[href='/view_cart']")
            )
        );

        ((org.openqa.selenium.JavascriptExecutor) driver)
            .executeScript("arguments[0].scrollIntoView(true);", viewCart);

        ((org.openqa.selenium.JavascriptExecutor) driver)
            .executeScript("arguments[0].click();", viewCart);

        System.out.println("Automation Exercise product added to cart");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://practicetestautomation.com/practice-test-login/");

        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");

        driver.findElement(By.id("password")).sendKeys("Password123");

        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
}
```

Open *App.java ~/DevOps/MyMavenFirefoxApp/src/main/java/com/example Save

```java
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();
        WebDriver driver = new FirefoxDriver(options);
        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920, 1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));
        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();
        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);
        driver.get("https://automationexercise.com/products");
        driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");
        driver.findElement(By.id("submit_search")).click();

        WebElement product = wait.until(
                ExpectedConditions.visibilityOfElementLocated(By.cssSelector("a[data-product-id='2']"))
        );

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView({block:'center'});", product);
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", product);
```

Java Tab Width: 8 Ln 28, Col 34 INS

```java
        WebElement viewCart = wait.until(
                ExpectedConditions.elementToBeClickable(By.cssSelector("#cartModal a[href='/view_cart']"))
        );
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView(true);", viewCart);
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", viewCart);

        System.out.println("Automation Exercise product added to cart");

        driver.switchTo().newWindow(WindowType.TAB);
        driver.get("https://practicetestautomation.com/practice-test-login/");
        Thread.sleep(2000);
        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();
        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);
        driver.quit();
    }
}
```

Java Tab Width: 8 Ln 28, Col 34 INS

Step 7: Build the Maven Project

Compile and build the project using the command:

mvn clean install

```
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenFirefoxApp >--------------------
[INFO] Building MyMavenFirefoxApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenFirefoxApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenFirefoxApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenFirefoxApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ MyMavenFirefoxApp ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 11] to target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.788 s
[INFO] Finished at: 2026-03-17T22:34:45+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$
```

Step 8: Run the Application

Run the application using:

mvn exec:java

```
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenFirefoxApp$ mvn exec:java
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenFirefoxApp >--------------------
[INFO] Building MyMavenFirefoxApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenFirefoxApp ---
```